

for datasets that include image annotations, such as Mathverse (Zhang et al., 2024a). Future work could explore how VC-CoT adapts to real-world images, potentially leveraging other types of annotations, such as pointing data from Pixmo (Deitke et al., 2024) for example, to bridge the gap between synthetic and real-world settings.

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A Model Selection and Details

We evaluate a diverse set of multimodal language models (MLLMs) spanning different architectures, vision encoders, and parameter scales. Table 8 provides details on the open-source models used in our experiments. These models are sourced from Hugging Face, with their specific repository paths listed for reproducibility.

In addition to open-source models, we also evaluate proprietary models using the OpenAI API. Specifically, we test gpt-4o and gpt-4-turbo, for comparison against open-source alternatives. Since their exact architectures and vision encoders are not publicly disclosed, we cannot report model details.

B Image Generation

B.1 One-Shape Images

To create a diverse dataset of geometric shapes, we generate images of squares, equilateral triangles, pentagons, hexagons, heptagons, octagons, and circles. Each shape is rendered in various sizes, orientations, and colors. Specifically, we vary the background color (white, black, red, blue) and fill color, selecting from a palette of 20 distinct hues. We ensured that when the background color equaled the shape color, those samples were not used. Additionally, each shape is rotated in steps of 10 degrees, covering a full 360-degree range. We account for shape symmetry, so we do not have duplicate samples on equivalent rotation angles (e.g., 90 degrees and 180 degrees when rotating a square). Finally, we vary in 9 different shape sizes. For the results in Table 1, we randomly sample a subset of **2,000** images to ensure diversity while maintaining computational efficiency. Figure 6 shows examples from our dataset.

B.2 Two-Shape Images

In addition to single-shape images, we generate images containing two distinct geometric shapes. Each image consists of a randomly selected pair from squares, equilateral triangles, pentagons, hexagons, heptagons, and octagons, placed in non-overlapping positions on a 400×400 canvas. Shapes vary in size, rotation, and color, with sizes randomly chosen from $\{80, 90\}$ pixels and rotations sampled in 10° increments. Each shape is assigned a fill color from a set of 20 hues, while the background color is randomly selected from white, black, red, or blue. In total, we generated **5,090** unique images, and used all of them for evaluation.

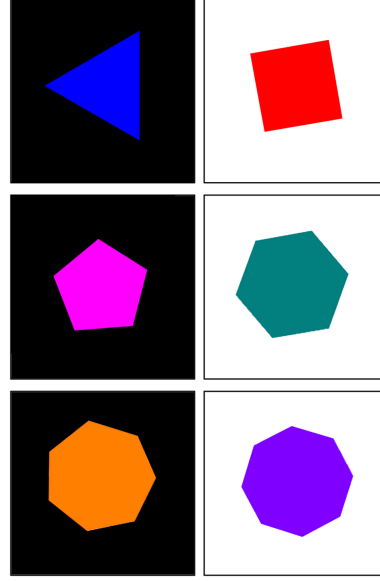


Figure 6: Examples of regular polygons (triangle through octagon) used in our evaluation in Section 3.1.

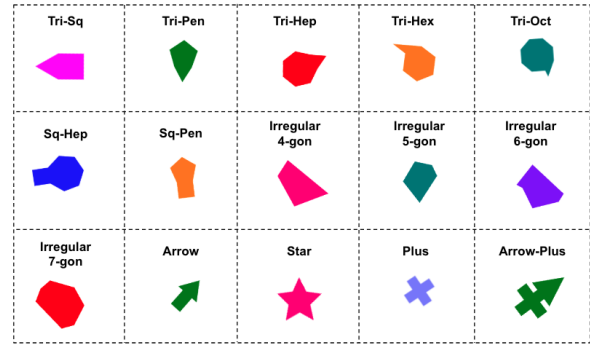


Figure 7: Examples of all merged, irregular and abstract shapes used to create Table 5

B.3 Abstract Shape Image Generation

Merged shapes. For the results in Table 5, we generate images containing two connected geometric shapes. We consider ten unique shape combinations, including Triangle-Square, Pentagon-Square, Hexagon-Triangle, and Octagon-Square, ensuring a range of structural complexities. Each combination is assigned a total side count based on the sum of its components. Shapes are placed in a connected configuration, rotated at random angles, and rendered with diverse colors on varying background shades. In total, **100** images for the merged shape category.

Irregular Polygons. We generate images of abstract irregular polygons with between 5 and 8 sides. Each polygon is randomly positioned on a 400×400 canvas with a randomly assigned background color (white, black, red, or blue) and a